

In class project

1. Choose suitable material of construction for 50 mm pipe operating at 2 bar pressure
 - 98% w/w sulfuric acid at 70C
 - 5% aqueous sodium hydroxide at 30C
 - 10% w/w hydrofluoric acid
 - 5% w/w sulfuric acid at 30C
 - 5% w/w nitric acid at 30C
2. Stainless steels are preferred in biological processes to reduce contamination by corrosion products. Nowadays, most fermentation plants are built from 316 stainless steel or 304 stainless steel. Decide which of these materials to use to replace the pipe of fermentation plant if the expected future life of the plant is 15 years.

- Stainless steel 316 pipe (inside diameter 4.026 inches and outside diameter 4.5 inches) which corrodes to the half in 7 years and needs to be changed.
- Thicker Stainless steel 316 pipe (inside diameter 3.5 inches and outside diameter 4.5 inches)
- Stainless steel 304 pipe (inside diameter 4.026 inches and outside diameter 4.5 inches) of which corrosion rate is 90% of stainless steel 316

Estimated cost for per unit length pipe is; Stainless steel 316 pipe \$24, Thicker Stainless steel 316 pipe \$28, Stainless steel 304 pipe \$26. (Installation and fitting for all materials is \$20 per unit length.)

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Chemical Engineering Design

98% w/w sulfuric acid at 70C

- High silicon cast iron

5% w/w sulfuric acid at 30C

- Lead

5% w/w nitric acid at 30 C

- stainless steel 18/8

5% aqueous sodium hydroxide at 30C

- PVC

10% hydrofluoric acid

- Lead

2.

a.) You change the pipe every 7 years so you need 3 SS 316 pipes. $3 \times \$24 = \72

b.) for corrosion rate of SS 316 from first problem $4.5 - 4.026 = 0.474$, $0.474 / 2 / 2 / 7 = 0.0169$ IPY
For half of the thicker SS 316 pipe is 0.25 in so $0.25 \text{ in} / 0.0169 \text{ IPY} = 14.76$ years to change so you need to buy 2 pipes. So $2 \times \$28 = \56

c.) Corrosion rate of SS 304 is $0.0169 \times 0.9 = 0.015$ IPY and the thickness is of half a pipe is 0.1185 in so $0.1185 \text{ in} / 0.015 \text{ IPY} = 7.9$ years so we only need 2 of the SS 304 pipe $2 \times \$26 = \52

So the best option would be option 3.